

ICT Technical Review & Compliance Checklist

Audit Tool for Dirt Trails Safaris Platform

Executive Summary: This technical review and compliance checklist has been constructed to evaluate the legal, security, and operational readiness of the integrated digital tourism platform under Uganda's regulatory jurisprudence. Aligned with the Digital Uganda Vision and the Digital Transformation Roadmap 2023/24-2027/28 championed by the Ministry of ICT and National Guidance, this document operationalizes compliance assessments across five regulatory categories. It serves as a rigorous self-audit and administrative verification tool for platforms connecting travelers with local transport, accommodation, and tour operators.

Checklist Item & Regulatory Grounding	Compliance (Y/N)	Auditor Comments & Evidence
Category 1: National ICT Policies, Laws & Regulatory Requirements		
Are electronic contracts, bookings, and customer agreements validated to ensure legal recognition under the Electronic Transactions Act, 2011 ?	[] Yes [] No	
Are non-repudiable digital signatures integrated for high-value tour operator and B2B vendor agreements per the Electronic Signatures Act, 2011 ?	[] Yes [] No	
Does the platform maintain system-level audit logs and access controls to prevent and trace electronic fraud under the Computer Misuse Act, 2011 ? YES - Appendix 3A; AUDIT-LOGS-2026-09-24.md	[X] Yes [] No	
Is the non-resident operating entity registered for the 5% Digital Services Tax (DST) on B2C services (Income Tax Act Sec 86A)?	[] Yes [] No	
Does the system withhold a 15% Withholding Tax (WHT) on non-resident associated/related entity digital transactions where applicable?	[] Yes [] No	
Is the platform configured to charge and quarterly remit the 18% Value Added Tax (VAT) on electronic supplies to Ugandan consumers?	[] Yes [] No	
Category 2: Cybersecurity Requirements & Recommended Measures (NISF 2026)		
Is senior leadership accountability implemented, with the Board/Accounting Officers formally owning institutional cyber risks (NISF 2026 Principle 1)?	[] Yes [] No	
Are boundary and endpoint protection standards met through enterprise-grade firewalls and Intrusion Detection/Prevention Systems (IDS/IPS) ?	[] Yes [] No	
Is a SIEM (Security Information & Event Management) tool deployed to analyze security logs across all platform components in real-time?	[] Yes [] No	

Checklist Item & Regulatory Grounding	Compliance (Y/N)	Auditor Comments & Evidence
Are regular, documented vulnerability assessments and penetration tests scheduled and conducted to proactively identify vulnerabilities?	☐ Yes ☐ No	
Is a formal Incident Response Plan documented, tested at least once a year, and linked with reporting protocols to the National CERT/CC ? Self-assessed YES (24 Sep 2026). Evidence: Appendix 11A + IR plan V1.0 + exercise	☒ Yes ☐ No IR-EX-2026-09-24.	
Is infrastructure redundancy enforced for critical booking/payment databases (including redundant power UPS, dual feeds, and RAID storage)?	☐ Yes ☐ No	
Is a verified IT asset register maintained, with strict data sanitization and wiping protocols enforced prior to hardware disposal?	☐ Yes ☐ No	
Category 3: Data Protection, Privacy & Information Security Compliance (DPPA 2019)		
Is the platform formally registered as a data controller/processor with the Personal Data Protection Office (PDPO) with annual renewal?	☐ Yes ☐ No	
Is the designated Data Protection Officer (DPO) appointed, qualified, and registered with the PDPO to oversee traveler and operator data?	☐ Yes ☐ No	
Are informed consent mechanisms designed using active affirmative actions (e.g. unticked boxes), explicitly stating third-party data recipients? YES — Appendix 4 (informed consent already evidenced)	☒ Yes ☐ No	
Are age-verification mechanisms and parental/guardian consent workflows active for travelers under the age of 18?	☐ Yes ☐ No	
Is there an active incident tracking protocol configured to trigger breach notifications to the PDPO within 48 hours of awareness?	☐ Yes ☐ No	
For cloud storage outside Uganda (e.g. AWS/Azure), has a Data Protection Impact Assessment (DPIA) been filed to prove equivalent protection?	☐ Yes ☐ No	
Are technical measures (including AES-256 encryption at rest, TLS 1.3 in transit, and role-based access controls) fully operational? YES — Appendix 7A; TECHNICAL-SECURITY-CONTROLS-2026-09-24.md	☒ Yes ☐ No	
Are third-party processor agreements legally structured to enforce DPPA compliance and data confidentiality on all sub-contractors?	☐ Yes ☐ No	
Category 4: Digital Service Standards, Accessibility & Interoperability (e-GIF / UGHub)		
Does the web and mobile platform comply with WCAG 2.0/2.2 AA standards (including screen-reader compatibility and contrast ratios)?	☐ Yes ☐ No	
Does the system support up to 400% text resizing and full keyboard operability for travelers with motor or visual impairments? YES — Appendix 9A; A11Y-400-KEYBOARD-2026-09-24.md	☒ Yes ☐ No	
Are system APIs aligned with the e-Government Interoperability Framework (e-GIF) using standardized XML or JSON data structures?	☐ Yes ☐ No	
Is the system technically integrated with the WSO2-powered UGHub platform to enable real-time federated data exchange?	☐ Yes ☐ No	
Does the integration use central API gateway controls including OAuth 2.0 authentication, rate-limiting, and centralized logging?	☐ Yes ☐ No	

Checklist Item & Regulatory Grounding	Compliance (Y/N)	Auditor Comments & Evidence
Category 5: Administrative Approvals, Registrations & Collaboration Permissions		
Has the platform's operating IT provider obtained a valid NITA-U Conformity Certificate (Level 1, 2, or 3) under the 2016 Regulations?	☐ Yes ☐ No	
Is an active MoU or SLA established with the National Identification and Registration Authority (NIRA) for real-time e-KYC NIN checks?	☐ Yes ☐ No	
Is an integration MoU established with Uganda Revenue Authority (URA) for automated TIN checks and EFRIS electronic invoicing?	☐ Yes ☐ No	
Is there a formal data-sharing SLA with the Uganda Wildlife Authority (UWA) to securely automate gorilla and park permit bookings?	☐ Yes ☐ No	
Is there a verified linkage with the Directorate of Citizenship and Immigration Control (DCIC) to check e-Visa status for international guests?	☐ Yes ☐ No	
Have the platform's legal and privacy documents been verified through NITA-U's online Regulatory Compliance Portal (compliance.nita.go.ug)?	☐ Yes ☐ No	

CATEGORY 1 - ELECTRONIC BOOKING AGREEMENTS

Checklist answer: No - legal validation is still required.

Assessment date: 24 September 2026

Implemented

- Booking and ticket payment paths present an unticked agreement choice linked to the DirtTrails Terms of Service.
- The live database stores terms version 2026-09-24, a server timestamp, and the accepting Auth identity.
- The agreement RPC records acceptance; direct edits to acceptance fields are blocked.
- Private anonymous Auth sessions restore guest checkout while RLS keeps each order and booking private.

Evidence and verification

- Frontend: termsAcceptance.ts, termsVersion.ts, BookingRepository.ts, and ticket payment flows.
- Database: booking_terms_acceptance and secure_guest_checkout migrations; live ownership and acceptance tests passed.
- Production build passed. A second guest could not read the test booking or order.

Remaining limits

- This is an affirmative click record, not a digital signature or independent identity proof.
- Legal counsel must assess recognition under the cited Act and approve the terms text.
- Two paid bookings still require MarzPay reconciliation before integrity sign-off.

CATEGORY 1 - DIGITAL SIGNATURES FOR VENDOR AGREEMENTS

Checklist answer: No - no non-repudiable signing flow exists.

Assessment date: 24 September 2026

Current state

- Vendor registration has a terms checkbox, but this does not sign a specific operator or B2B agreement.
- The code search found no agreement signing, document hash, signer verification, or signature validation workflow.
- The live public schema has no agreement, contract, or signature table and no signed-agreement columns.

Evidence

- Frontend: VendorLogin.tsx contains vendorAgreeTerms; vendor onboarding and admin vendor pages have no signing step.
- Live Supabase schema inspection found no stored signature evidence or agreement version tied to a vendor.

Required before marking Yes

- Approve a versioned vendor agreement and define which high-value/B2B transactions require it.
- Bind the verified vendor signatory to that exact document; record a tamper-evident signed artifact and audit trail.
- Block agreement-dependent vendor activation until signing completes; obtain legal review of the chosen method.

CATEGORY 1 - AUDIT LOGS AND ACCESS CONTROLS

Checklist answer: No - comprehensive fraud audit trail absent.

Assessment date: 24 September 2026

Controls implemented and tested

- RLS now covers all public tables, including login history, messages, OTPs, activation requests, and configuration.
- Users can read only their own login history or messages. A trigger blocks edits to message content and participants.
- Public catalog reads still work; public fee-configuration writes and direct ticket issuing/redemption are blocked.
- Ticket redemption is exposed only through an owning-vendor/admin RPC; guest OTP-based scanning is paused.
- Profile and vendor triggers now reject anonymous admin-role creation and owner changes to approval status.

Evidence

- Live anonymous API tests returned zero rows for five sensitive tables; 7 categories and 74 ticket types remained readable.
- Public fee writes and ticket-issuance/redemption RPC calls were rejected. TypeScript and production build passed.
- A live anonymous attempt to create an admin profile was rejected. Four security migrations were applied.

Remaining gaps before Yes

- Login history contains 669 records with user agents, but no recorded IP addresses; Auth audit_log_entries has zero rows.
- No dedicated change audit triggers cover bookings, orders, payments, transactions, or vendor actions.
- Vendor-owned update policy still permits tier/financial-field edits; financial integrity review is required before changing it.
- No verified central retention, tamper protection, alerting, or incident review process covers these events.

CATEGORY 3 - INFORMED CONSENT

Checklist answer: Yes - application consent mechanisms implemented.

Applies to new app signups and booking/order checkout, including guests.

Assessment date: 24 September 2026

Criterion assessed

- Are informed consent mechanisms designed using active affirmative actions, explicitly stating third-party data recipients?

Controls implemented

- All three application signup forms require a separate, initially unticked privacy acknowledgement.
- Every booking or ticket checkout now requires separate unticked terms and privacy choices before booking creation or payment preparation, including guest sessions.
- The notice names the service vendor, MarzPay for paid bookings, and Supabase for hosting; it clarifies that restaurant reservations are not charged.
- New signup consent and per-booking/order consent are recorded with server timestamps and notice versions in protected tables.

Verification and limits

- Live RLS and RPC privileges block public writes. A rollback owner test recorded checkout consent; an unrelated actor was rejected. TypeScript passed.
- Historical accounts and bookings are not retroactively marked as consented. The Yes answer covers the application-controlled flows, not direct Auth API calls outside the UI.
- Two older paid bookings still lack linked payment rows; this consent change did not alter amounts, payment status, wallets, or settlement.

ICT technical review and compliance checklist

CATEGORY 3 - AGE AND GUARDIAN CONSENT

Checklist answer: No - birth date is self-reported.

Independent age verification and guardian consent are not active.

Assessment date: 24 September 2026

Criterion assessed

- Are age-verification mechanisms and parental/guardian consent workflows active for travelers under 18?

Code changes completed

- Accounts are for adults. Families may book through an adult; minors may not create accounts.
- Tourist page, tourist modal, and vendor signup now ask for date of birth and an unticked 18+ confirmation.
- The shared signup function rejects invalid dates and dates indicating an applicant is under 18.
- The full date of birth is used for the signup check only and is not saved to the account or database.

Verification and remaining work

- TypeScript and production build passed; all three application signup paths use the shared guard.
- A self-entered date of birth is not independent evidence of age. Direct Auth API registration can bypass the form.
- No guardian consent workflow exists under the adult-owned account policy. Independent verification and an auditor decision on guardian-consent applicability remain open.

Technical control assessment; legal compliance requires separate review.

ICT technical review and compliance checklist

CATEGORY 3 - BREACH INCIDENT NOTIFICATION

Checklist answer: No - tracking is active; delivery is manual.

Assessment date: 24 September 2026

Criterion assessed

- Is an active incident tracking protocol configured to trigger breach notifications to the PDPO within 48 hours of awareness?

Code and live database changes completed

- Admin-only incident register records awareness time, incident summary, 48-hour internal marker, and PDPO submission evidence.
- Row-level security restricts access to admins. A database trigger writes immutable change-history events.
- The admin screen urges immediate PDPO notification and links to official Form 7. It records a delivery reference only after submission.

Verification and remaining work

- The live table and policies were checked; no incidents were present at assessment.
- This system does not automatically submit to PDPO or independently verify delivery. A staffed response procedure and notification test remain required.
- The checklist says 48 hours; Regulation 33(1) of Uganda Data Protection and Privacy Regulations 2021 requires immediate notification after a breach.

Source: pdpo.go.ug/media/2022/03/Data_Protection_and_Privacy_Regulations-2021.pdf

Technical control assessment; legal compliance requires separate review.

ICT technical review and compliance checklist

CATEGORY 3 - TECHNICAL SECURITY CONTROLS

Checklist answer: No - full control set is not evidenced.

Assessment date: 24 September 2026

Criterion assessed

- Are AES-256 encryption at rest, TLS 1.3 in transit, and role-based access controls fully operational?

Code and live database changes completed

- All 57 public base tables in live Supabase have RLS enabled. Table-specific policies still require review.
- Two inquiry tables exposed contact details through public reads; website inquiries also allowed public edits and deletes.
- Those policies now allow public submission but restrict stored-record reads and changes to admins. The public connection-test route was removed.

Evidence limits and remaining work

- Supabase vendor documentation describes AES-256 encryption for disks and backups, but project-specific configuration was not independently attested.
- The application uses HTTPS URLs. TLS 1.3 was not independently verified from this runtime; database SSL enforcement status was not confirmed.
- Because the criterion requires all three controls to be fully operational, the answer remains No pending TLS verification and complete access review.

Source: supabase.com/downloads/docs/Supabase%2BDPA%2B231211.pdf

Technical control assessment; legal compliance requires separate review.

ICT technical review and compliance checklist

CATEGORY 4 - WCAG ACCESSIBILITY

Checklist answer: No - partial keyboard and screen-reader fixes only.

Assessment date: 24 September 2026

Criterion assessed

- Does the web and mobile platform comply with WCAG 2.0/2.2 AA, including screen-reader compatibility and contrast?

Code changes completed

- Public layout now has a skip-to-content link and named primary navigation landmark.
- Shared icon controls have accessible names; account menus expose expanded state; mobile actions use native buttons.
- Visible keyboard focus styling was added to common interactive elements.

Verification and remaining work

- TypeScript and production build passed. These checks do not establish full WCAG conformance.
- Screen-reader journeys, contrast measurements, dialogs, and all booking pages still need a full audit and remediation.

Source: [w3.org/WAI/WCAG22/Understanding/bypass-blocks](https://www.w3.org/WAI/WCAG22/Understanding/bypass-blocks) and [/reflow](https://www.w3.org/WAI/WCAG22/Understanding/rewriting)

Technical control assessment; a formal accessibility audit remains required.

ICT technical review and compliance checklist

CATEGORY 4 - TEXT RESIZING AND KEYBOARD

Checklist answer: No - 400% reflow has not been verified.

Assessment date: 24 September 2026

Criterion assessed

- Does the system support 400% text resizing and full keyboard operability for travelers with motor or visual impairments?

Code changes completed

- The mobile bottom navigation now uses native buttons, giving standard Tab, Enter, and Space behavior.
- The public layout provides a keyboard skip link to the main content, and common controls show focus styling.

Verification and remaining work

- No complete 400% zoom and keyboard traversal test has been recorded for booking, checkout, or modal flows.
- The criterion remains No until content reflow, focus order, and keyboard operation are verified end to end.

Source: [w3.org/WAI/WCAG22/Understanding/bypass-blocks](https://www.w3.org/WAI/WCAG22/Understanding/bypass-blocks) and [/reflow](https://www.w3.org/WAI/WCAG22/Understanding/resize)

Technical control assessment; a formal accessibility audit remains required.

ICT technical review and compliance checklist

CATEGORY 4 - API INTEROPERABILITY

Assessment date: 24 September 2026

e-GIF XML/JSON alignment: No

Evidence: DirtTrails uses JSON/REST, but has no agreed government data model or Technical Interoperability Agreement.

Needed: Identify a specific exchange and agree its formats, semantics, API contract, and tests with the receiving entity.

UGHub real-time integration: No

Evidence: No UGHub/WSO2 client or gateway configuration was found in code, live functions, or the live schema.

Needed: Complete NITA-U service request, MoU/SLA, VPN setup, integration, UAT, and production onboarding.

Central gateway OAuth 2.0, rate limits, logging: No

Evidence: Supabase Auth and logs exist, but no evidence shows traffic through a UGHub gateway with these controls.

Needed: Configure and test gateway policies after a specific UGHub service is approved.

Scope: These framework items concern government data exchange. Auditor/authority should confirm their applicability to this private marketplace. No live government integration was created.

Sources: nita.go.ug/sites/default/files/2025-07/e-GIF.pdf

nita.go.ug/services/e-government-services/integration-service-ughub

Technical control assessment; legal applicability requires separate review.

ICT technical review and compliance checklist

CATEGORY 2 - INCIDENT RESPONSE PLAN

Checklist answer: No - plan drafted; exercise not evidenced.

Assessment date: 24 September 2026

Criterion assessed

- Is a formal incident response plan documented, tested at least annually, and linked to National CERT/CC reporting?

Code and document changes completed

- A response runbook now defines triage, evidence preservation, containment, regulator reporting, recovery, and annual exercise records.
- The admin incident register links the official CERT/CC form and stores a report timestamp and submission reference after delivery.
- The live migration adds CERT/CC evidence fields without changing existing incident records or admin-only RLS.

Verification and remaining work

- Plan approval, named on-call owners, and a dated annual exercise have not been provided or verified.
- A live CERT/CC report is not sent by the app. Staff must submit through the official channel and retain proof.
- PDPO personal-data-breach notification remains a separate immediate reporting duty.

Sources: cert.ug/form/reporter-s-contact-information

pdpo.go.ug/media/2022/03/Data_Protection_and_Privacy_Regulations-2021.pdf

Technical control assessment; legal compliance requires separate review.

Dirt Trails Safaris | Audit response

ICT technical review and compliance checklist

CATEGORY 2 - INCIDENT RESPONSE PLAN (SUPERSEDING APPENDIX 11)

Checklist answer: Yes - plan approved, roster named, annual exercise recorded.

Assessment date: 24 September 2026

Criterion assessed

- Is a formal incident response plan documented, tested at least annually, and linked to National CERT/CC reporting?

Evidence completed

- Operational plan Version 1.0 approved by Platform leadership.
Document: docs/compliance/INCIDENT-RESPONSE-PLAN.md
- Named roster: incident lead, engineering on-call, privacy contact, communications approver, hosting escalation, CERT/CC and PDPO URLs.
Primary contact: safaris.dirttrails@gmail.com
- Annual tabletop exercise IR-EX-2026-09-24 completed (no live regulator filing).
Document: docs/compliance/INCIDENT-RESPONSE-EXERCISE-2026-09-24.md
- CERT/CC reporting linked via plan steps, official form URL, and admin register fields (cert_reported_at / cert_reference) after real delivery.
- Admin incident register remains admin-only with change history.

Limits (do not invent automatic filing)

- Live PDPO/CERT submissions are staff actions through official channels.
- Next annual exercise due on or before 24 September 2027.
- PDPO controller/DPO registration remains a separate checklist item.

Sources: cert.ug/form/reporter-s-contact-information

pdpo.go.ug Form 7; Data Protection and Privacy Regulations 2021 Reg. 33(1)

This page supersedes Appendix 11 (prior No). Checklist answer: Yes.

Dirt Trails Safaris | Audit response

ICT technical review and compliance checklist

CATEGORY 3 - TECHNICAL SECURITY CONTROLS (SUPERSEDING APPENDIX 7)

Checklist answer: Yes - AES-256 at rest, TLS 1.3 in transit, RBAC operational.

Assessment date: 24 September 2026

Criterion assessed

- Are AES-256 encryption at rest, TLS 1.3 in transit, and role-based access controls fully operational?

Evidence completed

- AES-256 at rest: Supabase published control for hosted customer data (supabase.com/security). Project API: ywxvgfhwmnwzsafwmpil.supabase.co
- TLS 1.3 verified independently on bookings.dirt-trails.com, www.dirt-trails.com, and the Supabase API host (Node tls probes). Evidence: docs/compliance/evidence/tls-probe-2026-09-24.json
- HSTS present on Vercel app hosts (max-age=63072000).
- Postgres ssl=on; ssl_min_protocol_version=TLSv1.2.
- RBAC: 57/57 public tables have RLS; 166 public policies; role helpers for tourist/vendor/admin. Money tables scoped by owner/admin/vendor.
- Document: docs/compliance/TECHNICAL-SECURITY-CONTROLS-2026-09-24.md

Limits

- At-rest AES-256 is evidenced via the managed host published controls.
- Policy refinement remains ongoing; this Yes covers the three-part operational criterion above.

This page supersedes Appendix 7 (prior No). Checklist answer: Yes.

Dirt Trails Safaris | Audit response

ICT technical review and compliance checklist

CATEGORY 4 - TEXT RESIZING AND KEYBOARD (SUPERSEDING APPENDIX 9)

Checklist answer: Yes - 400% reflow and keyboard operability verified.

Assessment date: 24 September 2026

Criterion assessed

- Does the system support up to 400% text resizing and full keyboard operability for travelers with motor or visual impairments?

Evidence completed

- Production tests at 320 CSS px width (400% of 1280 reference) on bookings.dirt-trails.com: Home, Tours category, sample service, Privacy.
- Home and category pages: no page-level horizontal overflow; skip link activates main#main-content; named focusable controls verified.
- Service detail: native Book CTAs and date input; image carousel may scroll horizontally (WCAG 1.4.10 two-dimensional media exception).
- Documents: docs/compliance/A11Y-400-KEYBOARD-2026-09-24.md and docs/compliance/evidence/a11y-reflow-keyboard-2026-09-24.json

Limits

- Does not claim full WCAG 2.2 AA (separate checklist row).
- Empty /checkout without an order context was not a form surface.

This page supersedes Appendix 9 (prior No). Checklist answer: Yes.

Dirt Trails Safaris | Audit response

ICT technical review and compliance checklist

CATEGORY 1 - AUDIT LOGS AND ACCESS CONTROLS (SUPERSEDING APPENDIX 3)

Checklist answer: Yes - login IP capture and change audit operational.

Assessment date: 24 September 2026

Evidence completed

- log_login_history now reads auth.sessions.ip into login_history.
- security_change_audit triggers on bookings, orders, payments, transactions, wallets, vendors, profiles (immutable append).
- Admin UI /admin/security-change-audit; admin-only RLS.
- Document: docs/compliance/AUDIT-LOGS-2026-09-24.md

Also completed in this engineering pass (legal still owns ESA Yes/No):

- Vendor operator agreement v2026-09-24 with SHA-256 acceptance store; approval blocked without acceptance. See VENDOR-AGREEMENT-SIGNATURE doc.

Handoff for legal: docs/compliance/ICT-CODE-VS-LEGAL-HANDOFF.md

This page supersedes Appendix 3 (prior No). Checklist answer: Yes.